

# HW2\_18

A life insurance company examines the relationship between the amount of life insurance held by a household and household income. Let  $INCOME$  be household income (thousands of dollars) and  $INSURANCE$  the amount of life insurance held (thousands of dollars). Using a random sample of  $N = 20$  households, the least squares estimated relationship is Given

Estimated regression:

$$\widehat{INSURANCE} = 6.855 + 3.880 \cdot INCOME$$

Standard errors:

$$se(\hat{\beta}_1) = 7.383, \quad se(\hat{\beta}_2) = 0.112$$

Sample size:  $N = 20$

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## (a) Sketch and Mean Point

Draw a sketch of the fitted relationship identifying the estimated slope and intercept. The sample mean of  $INCOME = 59.3$ . What is the sample mean of the amount of insurance held? Locate the point of the means in your sketch.

Sample mean of income:

$$INCOME = 59.3$$

```
# parameters
intercept <- 6.855
slope <- 3.880
income_mean <- 59.3

# mean of insurance
insurance_mean <- intercept + slope * income_mean

# create sequence for plotting
income_seq <- seq(0, 120, length.out = 100)
insurance_hat <- intercept + slope * income_seq

# plot
plot(income_seq, insurance_hat, type = "l",
     xlab = "Income (thousand dollars)",
     ylab = "Insurance (thousand dollars)",
     main = "Fitted Relationship between Income and Insurance")

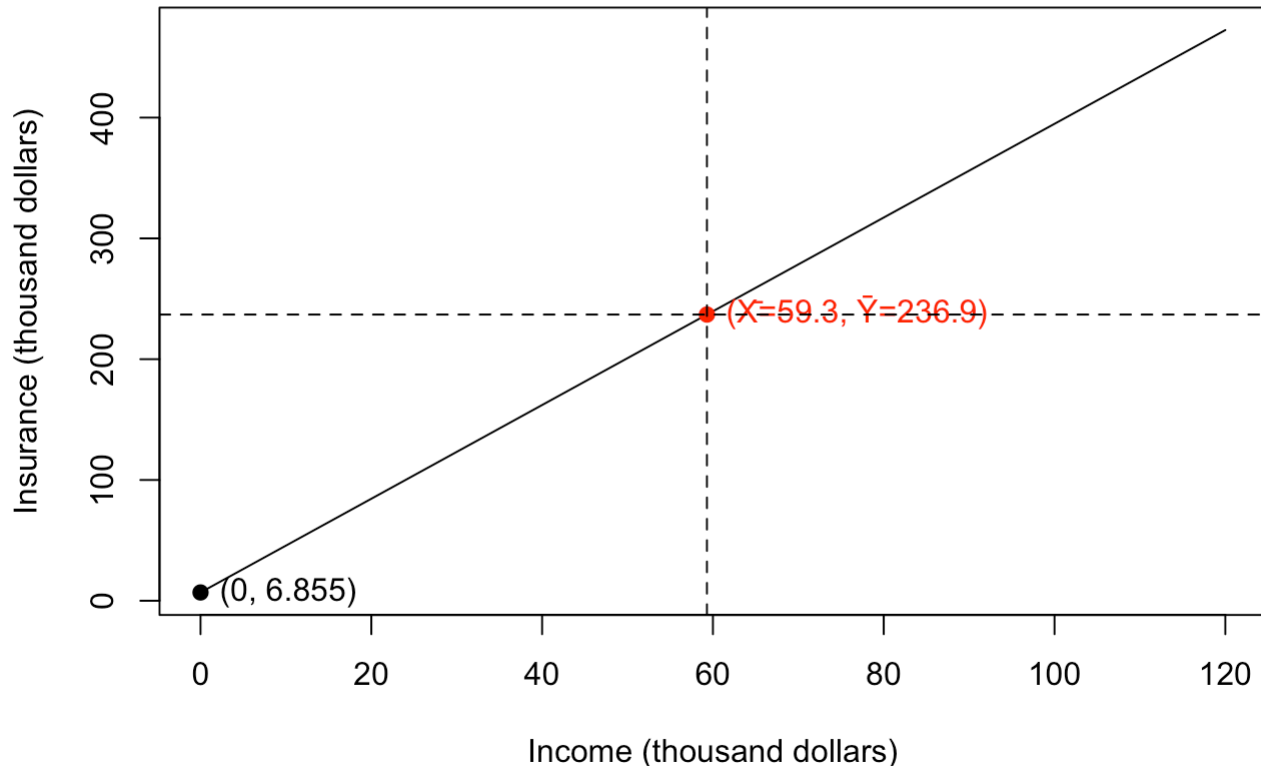
# add intercept point
points(0, intercept, pch = 19)
text(0, intercept, labels = "(0, 6.855)", pos = 4)

# add mean point (X_bar, Y_bar)
points(income_mean, insurance_mean, col = "red", pch = 19)
text(income_mean, insurance_mean,
     labels = paste0("(X̄=", round(income_mean,1),
                          ", Ȳ=", round(insurance_mean,1), ")"),
     pos = 4, col = "red")

# add vertical line at X mean
abline(v = income_mean, lty = 2)

# add horizontal line at Y mean
abline(h = insurance_mean, lty = 2)
```

## Fitted Relationship between Income and Insurance



### (b) Effect of \$1000 Increase in Income

How much do we estimate that the average amount of insurance held changes with each additional \$1000 of household income? Provide both a point estimate and a 95% interval estimate. Explain the interval estimate to a group of stockholders in the insurance company

Since INCOME is in thousands:

- A \$1000 increase → INCOME increases by 1
- Estimated change = 3.880 (thousand dollars)

$$CI = \beta_2 \pm t_{0.025, n-2} SE(\hat{\beta}_2)$$

```
beta2 <- 3.880
se_beta2 <- 0.112
t_crit_95 <- qt(0.975, df = 18)

lower <- beta2 - t_crit_95 * se_beta2
upper <- beta2 + t_crit_95 * se_beta2

c(lower, upper)
```

```
## [1] 3.644697 4.115303
```

For every \$1000 increase in income, customers tend to purchase about \$3880 more insurance. In repeated sampling, about 95% intervals constructed this way will contain the true value.

## (c) 99% Confidence Interval for INCOME = 100

Construct a 99% interval estimate of the expected amount of insurance held by a household with \$100,000 income. The estimated covariance between the intercept and slope coefficient is  $-0.746$ .

We estimate:

$$\hat{y} = 6.855 + 3.880 \times 100$$

```
x0 <- 100
var_b1 <- 7.383^2
var_b2 <- 0.112^2
cov_b1b2 <- -0.746

# Variance of prediction
var_yhat <- var_b1 + (x0^2)*var_b2 + 2*x0*cov_b1b2
se_yhat <- sqrt(var_yhat)

yhat <- intercept + slope * x0

t_crit_99 <- qt(0.995, df = 18)

lower <- yhat - t_crit_99 * se_yhat
upper <- yhat + t_crit_99 * se_yhat

c(yhat, lower, upper)
```

```
## [1] 394.8550 378.8936 410.8164
```

- The estimated expected amount of insurance is 394.8550.
- In repeated sampling, about 99% intervals constructed this way will contain the true value.

## (d) Hypothesis Test: $\beta_2 = 5$

One member of the management board claims that for every \$1000 increase in income the average amount of life insurance held will increase by \$5000. Let the algebraic model be  $\text{INSURANCE} = \beta_1 + \beta_2 \text{INCOME} + e$ .

Test the hypothesis that the statement is true against the alternative that it is not true. State the conjecture in terms of a null and alternative hypothesis about the model parameters. Use the 5% level of significance. Do the data support the claim or not? Clearly, indicate the test statistic used and the rejection region.

- Hypotheses

$$H_0 : \beta_2 = 5$$

$$H_1 : \beta_2 \neq 5$$

```
beta2_hat <- 3.880
beta2_null <- 5
se_beta2 <- 0.112

t_stat <- (beta2_hat - beta2_null) / se_beta2
t_stat
```

```
## [1] -10
```

```
qt(0.975, df = 18)
```

```
## [1] 2.100922
```

Since  $|t| = -10 > t_{0.975,18} = 2.100922$ , we reject  $H_0$ . The data do NOT support the claim that insurance increases by \$5000 per \$1000 income.

### (e) Test: $\beta_2 = 1$ vs $\beta_2 > 1$

Test the hypothesis that as income increases the amount of life insurance held increases by the same amount. That is, test the null hypothesis that the slope is one. Use as the alternative that the slope is larger than one. State the null and alternative hypotheses in terms of the model parameters. Carry out the test at the 1% level of significance. Clearly indicate the test statistic used, and the rejection region. What is your conclusion?

- Hypotheses

$$H_0 : \beta_2 = 1$$

$$H_1 : \beta_2 > 1$$

```
beta2_null <- 1  
  
t_stat <- (beta2_hat - beta2_null) / se_beta2  
t_stat
```

```
## [1] 25.71429
```

```
qt(0.99, df = 18)
```

```
## [1] 2.55238
```

Since  $|t| = 25.71429 > t_{0.99,18} = 2.55238$ , we reject  $H_0$  and conclude that the slope is significantly greater than 1.